COURSE SYLLABUS IN EE 211 – ELECTRICAL CIRCUITS 1

PART I. UNIVERSITY INFORMATION

VISION	Premier research university transforming communities in the ASEAN and beyond.
MISSION	USeP shall contribute to inclusive growth and sustainable development through continual innovation in proactive academic programs, impactful research for utilization, economically empowering community services, and sustainable resource management.
GOALS	● Internationalized University programs and services. ● Innovation and RDE-nurturing environment. ● Proactive, globally competitive, and service-oriented graduates. ● Transformed vulnerable and disadvantaged groups and communities. ● Mobilized and optimized resources for sustainable development.
EDUCATIONAL PHILOSOPHY	
Nature of Education in the University	The University of Southeastern Philippines is a state university that contributes to the development of human capital and to fostering innovations to transform communities in Mindanao, the Filipino nation, and the world. As a community of scholars, we aim to develop life-long learners who, guided by conscience, actively engage in the pursuit of generating and extending knowledge through research, extension, and responsible professional practice.
Principles for Teaching and Learning in the University	In all our educational endeavors, we strive to be continually guided by the following principles: ● Teaching and learning is based on science and evidence stemming from experience and reflections on the teaching practice. ● A conducive learning environment is founded on integrity and mutual respect, where learning is a co-construction among scholars. ● The learner is an active participant in the learning process to develop oneself and others' potential. ● A transformative learning experience is grounded in critical and reflective thinking and provides opportunities for practice, improvement, and action.
INSTITUTIONAL GRADUATE ATTRIBUTES	
IGA1: LEADERSHIP SKILLS	Creates and inspires positive changes in the organization; exercises responsibility with integrity and accountability in the practice of one's profession or vocation.
IGA2: CRITICAL AND ANALYTICAL THINKING SKILLS	Demonstrates creativity, innovativeness, and intellectual curiosity in optimizing available resources to develop new knowledge, methods, processes, systems, and value-added technologies.
IGA3: SERVICE ORIENTED	Demonstrates concern for others, practices professional ethics, honesty, and exemplifies socio-cultural, environmental concern, and sustainability.
IGA4: LIFELONG LEARNING	Demonstrates enthusiasm and passion for continuous personal and professional development.
IGA5: PROFESSIONAL COMPETENCE	Demonstrates proficiency and flexibility in the area of specialization and in conveying information in accordance with global standards

PART II. PROGRAM INFORMATION

Program Name: Bachelor of Science in Electrical Engineering

PROGRAM GRADUATE OUTCOMES			
PGO	**Description**	**Performance Indicator**	**IGA Aligned to**
PGO a	Apply knowledge of mathematics and sciences to solve engineering problems	a.1. Apply concepts in mathematics to solve engineering problems.	IGA2
PGO e	Identify, formulate, and solve engineering problems	e.2. Apply concepts, theories, and principles in solving engineering problem(s)	IGA5

PART III. DOCUMENT INFORMATION

Version Number	:	2026–1.0
Version Note	:	✓ New ☐ Minor Revision ☐ Major Revision ☐ No Revision
Describe Revision *(if applicable)*	:	
Semester	:	1st Semester
Academic Year	:	2026-2027

PART IV. FACULTY INFORMATION

Faculty Name	:	ELWIN JAY J. TANO
Email Address	:	ejjtano0120241110@usep.edu.ph
Contact Number	:	09262995868
Office	:	College of Engineering
Consultation Hours	:	Monday-Friday, 8:00 AM – 05:00 PM (excluding lunch break)

PART V. COURSE INFORMATION

Course Code	:	EE 211
Course Title	:	Electrical Circuits 1
Pre-requisite	:	NAT SCI 2 – PHYSICS FOR ENGINEERS MATH 121 – ENGINEERING CALCULUS 2
Credit	:	3.0 Units – Lecture
Level	:	Second Year

SDG Alignment:

SDG	Description of Alignment
7: Affordable and Clean Energy	This course introduces the core concepts of electrical power, energy-storing elements (capacitors and inductors), and maximum power transfer. A rigorous understanding of voltage, current, and power calculation is the absolute prerequisite for designing, analyzing, and optimizing renewable energy systems, smart grids, and energy-efficient electrical networks in the future.
9: Industry, Innovation, and Infrastructure	Circuit analysis is the fundamental language of all electrical infrastructure and technological innovation. By learning to simplify complex multi-source networks and analyze transient and steady-state responses, students develop the technical competencies required to innovate and maintain resilient industrial power systems, electronics, and communication infrastructures.

Course Description:

The course covers nodal and mesh analysis; application of network theorems in circuit analysis; analysis of circuits with controlled sources and ideal op-amps fundamentals of capacitors and inductors; analysis of dc-driven RL, RC, and RLC circuits; sinusoidal steady-state analysis of general RLC circuits.

Workload of Students:

With **18 weeks** at **3 contact hours** per week (**54 contact hours total)** the expected student workload is as follows:

- Group Activities for Formative Assessment
- Total of 4 quizzes (2 for both Midterm and Final Term)
- Midterm and Final Examinations

In instances where face-to-face sessions are impractical, online sessions, asynchronous class, or their equivalent will be provided.

Course Learning Outcomes (CLOs)

On the completion of the course, the student is expected to be able to do the following:

Code	Course Learning Outcomes	PGO Aligned to
CLO1	Given DC and AC circuit networks, the student will calculate unknown circuit variables by accurately solving necessary linear and differential equations.	PGO a
CLO2	Given multi-source electrical networks and time-varying circuits, the student will determine equivalent circuits and complete electrical responses by correctly applying fundamental network theorems and circuit principles.	PGO e

PART VI. ASSESSMENT ACTIVITIES

Formative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
FA1	Group Activity	Collaborative, team-based projects or assessments where students analyze, calculate, and simulate fundamental DC and AC electrical networks (e.g., verifying Thevenin/Norton equivalent circuits, or evaluating RLC transient responses). This integrates theoretical mathematical modeling with circuit simulation software to reinforce core circuit analysis principles and develop foundational engineering problem-solving skills.	CLO1 CLO2

Summative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
SA1	Quizzes	Periodic summative assessments comprising conceptual questions and structured mathematical problem-solving scenarios. Designed to incrementally evaluate the students' analytical grasp of fundamental circuit laws, network theorems, energy-storing elements, and AC/DC circuit analysis techniques prior to major examinations.	CLO1, CLO2
SA2	Term Examination	Comprehensive summative assessments designed to evaluate the students' overall mastery of the course material. These proctored examinations require students to synthesize knowledge through a rigorous mix of conceptual theoretical questions, complex electrical network analysis, and multi-step mathematical problem-solving scenarios (e.g., nodal and mesh analysis, application of equivalent circuit theorems, RLC transient responses, and frequency-domain phasor analysis).	CLO1, CLO2

PART VII. MEASUREMENT SYSTEM

Assessment Activity: FA1: Group Activity Rubric

Area to Assess	Excellent (100%)	Proficient (80%)	Developing (60%)	Beginning (40%)	Poor (20%)	Unacceptable (0%)
Application of Concepts (40%)	Thorough, correct application of foundational circuit principles (e.g., Kirchhoff's laws, network theorems, transient/phasor analysis) throughout the task.	Mostly accurate use of circuit analysis concepts with minor theoretical or mathematical misunderstandings.	Partial or inconsistent application of core circuit principles; some equations or models are set up incorrectly.	Misapplied or completely misunderstood key circuit concepts, laws, and theorems.	Application is entirely incorrect, yielding completely invalid circuit models or equations.	No attempt to apply the concepts at all.
Team Collaboration (25%)	All members contributed equally; excellent communication, division of problem-solving tasks, and cooperation.	Most members contributed actively; good collaboration and communication evident.	Uneven participation among members; signs of limited coordination when solving the circuit problems.	Dominance by one or a few members; evident lack of communication or shared problem-solving.	Dysfunctional group dynamic; visible conflict or complete lack of coordination in tackling the assignment.	No coordination or teamwork at all.
Output Quality (25%)	Output is complete, mathematically accurate, and features neat, properly labeled circuit diagrams and/or simulation schematics.	Output is mostly complete and neat but contains minor calculation error, missing units, or slightly untidy diagrams.	Output is functional but contains noticeable errors, missing units, or poorly drawn circuit schematics.	Output is incomplete, inaccurate, or lacks required circuit diagrams and proper unit notation.	Output is severely deficient, illegible, or completely unprofessional.	No output submitted.
Analytical Approach (10%)	Demonstrates highly, logical, efficient, and well-structured mathematical approaches to simplify and solve complex circuit problem.	Use standard, logical, and step-by-step methods to solve the problem, though perhaps not the most efficient route.	The approach is somewhat disorganized, making mathematical logic and derivation difficult to follow.	Rely heavily on guesswork or lack a clear, step-by-step mathematical strategy to analyze the circuit.	The approach is entirely chaotic, showing no understanding of circuit analysis methodology.	No methodology shown (e.g., providing only final answers without the supporting work).

Assessment Activity: SA1/SA2: Problem Solving Rubric

Area to Assess	Excellent (100%)	Proficient (80%)	Developing (60%)	Beginning (40%)	Poor (20%)	Unacceptable (0%)
Problem Formulation & Methodology (40%)	Correctly identify the circuit problem type and select the appropriate laws, theories (e.g., KVL, KCL, Superposition), and mathematical methods (e.g., nodal/mesh matrices, phasor transforms).	Mostly correct formulation and selection of circuit theorems with only minor logical gaps or inefficiencies in equation setup.	Partial understanding of the circuit; selects some relevant laws (like Ohm's law) but misses key methodological steps for the specific theorem required.	Major conceptual errors in setting up the circuit problem; sets up incorrect nodal/mesh equations or applies the wrong network theorem.	Barely formulates the problem; relies on completely irrelevant concepts or random guessing of circuit relationships.	Does not attempt to formulate or set up the circuit problem.
Correctness and Accuracy (40%)	Flawless execution. The final numerical answer for circuit variables is completely accurate with correct units (e.g., V, A, W, rad/s) and computational precision (including complex numbers for AC).	Conceptually solid solution but contains minor computational mistakes, algebraic errors, or missing/incorrect units.	Noticeable computational gaps (e.g., matrix solving errors, incorrect transient equations) resulting in an only partially correct final answer.	Largely incorrect final answer due to major computational errors in solving the systems of equations or complex algebra.	Fundamentally incorrect answer with no valid mathematical steps leading to a solution.	Does not attempt to solve the computations.

Clarity and Organization (20%)	Steps are logically presented, and circuit schematics are clearly drawn and labeled (e.g., indicating current directions, voltage polarities, and nodes).	Mostly clear presentation with only slight issues in organization, notation, or schematic labeling.	Somewhat disorganized; reasoning between mathematical steps is unclear or equivalent circuit diagrams are missing essential labels.	Highly disorganized, messy, lacks logical flow, and missing required equivalent circuit diagrams entirely.	Illegible or completely unstructured, making it impossible to follow the logic or identify which part of the circuit is being analyzed.	No evidence of written steps or diagrams.

PART VIII. GRADING SYSTEM AND ASSESSMENT PLAN:

The final grade in this course will be composed of the following items and their weights in the final grade computation:

Assessment Item	Percentage of Final Grade	CLO1	CLO2
SA1	40%	Scores/Problem Solving Rubric	Scores/Problem Solving Rubric
SA2	60%	Scores/Problem Solving Rubric	Scores/Problem Solving Rubric
TOTAL	**100%**		

Transmutation	:	Base-50
Passing Grade	:	3.00
Passing Grade Conditions	:	Students shall take all assessments and examinations and must have an attendance of not less than 80% in the class. In case of lacking output or examination, a grade of INC (Incomplete) shall be given in accordance with the existing rules and regulations reflected in the student handbook. In case of incurring absences of more than 20% of the total expected number of attendances, a grade of UW (Unauthorized Withdrawal) will be reflected in the grade sheet.

Grade Equivalence Table:

< 75 = 5.00	75-76 = 3.00	77-79 = 2.75	80-82 = 2.50	83-85 = 2.25
86-88 = 2.00	89-91 = 1.75	92-94 = 1.50	95-97 = 1.25	98-100 = 1.00

PART IX. LEARNING PLAN

In order to achieve the outcomes of this course, learners will go through this learning plan:

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
• Recognize the university VMGO & Institutional Graduate Attributes. • Appreciates the significance of PEOs and PGOs of the programs. • Recognize the rules and regulations of the course and the grading system.		1 hour	• Course Orientation • VMGO & IGA of the University • PEOs & PGOs of the program. • Rules and regulations of the course & grading system • Presentation of the course outline	• On-site or online meeting	• UseP Student Handbook (2016 edition)	• N/A
• Formulate basic circuit equations using voltage and current laws to solve unknown variables in fundamental networks. (CDIO Skills: 1.2.1, 2.1.1)	CLO1	4 hours	• **DC/AC SOURCES AND ELECTRICAL CIRCUIT COMPONENTS, VOLTAGE AND CURRENT LAWS**	• Student-based Learning • Flipped Classroom • Collaborative and Team Learning • Active Learning • Technology Integration • Problem sets • Lecture	• Ref01-08	• FA1 – Group Activity • SA1 – Quiz
• Set up and solve linear equations using general nodal and mesh analysis methods for complex circuit networks. (CDIO Skills: 1.2.1, 2.1.2)	CLO1	7 hours	• **NODAL AND MESH ANALYSIS** o General mesh analysis o General nodal analysis			
• Simplify multi-source electrical networks into equivalent circuits using standard theorems. (CDIO Skills: 1.2.1, 2.1.3, 2.1.4)	CLO2	12 hours	• **CIRCUIT ANALYSIS TECHNIQUES** o Linearity and superposition o Source transformation o Thevenin and Norton equivalent circuits o Maximum power transfer o Delta-Wye conversion o Circuits with controlled sources and the ideal op amp			• FA1 – Group Activity • SA1 – Quiz
• Demonstrate mastery of foundational circuit analysis and network theorems.	CLO1 CLO2	3 hours	**MIDTERM EXAMINATION**	• **Conduct of Midterm Examination – Independent Assessment**		• **SA2 – Term Examination**
• Explain the characteristics of energy-storing elements and calculate their voltage-current relationship. (CDIO Skills: 1.2.1, 2.1.1)	CLO1	4 hours	• **CHARACTERISTICS OF ENERGY-STORING ELEMENTS** o Capacitors and capacitance o Inductors and inductance	• Student-based Learning • Flipped Classroom • Collaborative and Team Learning • Active Learning • Technology Integration • Problem Sets • Lecture	• Ref01-08	• FA1 – Group Activity • SA1 – Quiz
• Determine the transient and steady-state responses of source-free and driven first-order circuits. (CDIO Skills: 1.2.1, 2.1.2, 2.1.3)	CLO1 CLO2	8 hours	• **ANALYSIS OF RL AND RC CIRCUITS** o Source-free RL and RC circuits o Driven RL and RC circuits		• Ref01-08	
• Calculate the complete response of series and parallel second-order	CLO1 CLO2	6 hours	• **ANALYSIS OF RLC CIRCUITS** o Source-free series and parallel RLC circuits	• Student-based Learning • Flipped Classroom		• FA1 – Group Activity

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
circuits. (CDIO Skills: 1.2.1, 2.1.2, 2.1.5)			○ Complete response of RLC circuits	• Collaborative and Team Learning • Active Learning • Technology Integration • Learning Journals • Lecture		• SA1 – Quiz
• Apply phasor concepts and impedance parameters to perform nodal and mesh analysis in the frequency domain. (CDIO Skills: 1.2.1, 2.1.2, 2.1.4)	CLO1 CLO2	6 hours	• **SINUSOIDAL STEADY-STATE ANALYSIS IN THE FREQUENCY DOMAIN** ○ The phasor concept and phasor diagram ○ Concept of impedance and admittance ○ Nodal and mesh analysis			
• Demonstrate mastery of time-varying circuits, transient analysis, and frequency-domain steady-state analysis.	CLO1 CLO2	3 hours	**FINAL EXAMINATION**	• **Conduct of Final Examination – Independent Assessment**		• **SA2 – Term Examination**

PART X. REFERENCES/ARTICLES TO READ

Code	Reference/Article
Ref01	Alexander, C.K. & Sadiku, M.N.O. (2020). Fundamentals of Electric Circuits (7th ed.). McGraw Hill
Ref02	Grob, M. (2021). Grob's Basic Electronics (13th ed.). McGraw Hill
Ref03	Hayt, W.H., Kemmerly, J.E., & Durbin, S.M. (2019). Engineering Circuit Analysis. Ninth edition. McGraw-Hill Education, New York, NY
Ref04	Tan, J.C. (2004). Electric Circuit Analysis. National Bookstore
Ref05	Nahvi, M. & Edminister. J.A. (2003). Schaum's Outline of Theory and Problems of Electric Circuits. Fourth edition. McGraw-Hill
Ref06	Gusson, M. (2007). Schaum's Outline of Basic Electricity. Second edition. McGraw-Hill
Ref07	Nahvi, M. & Edminister. J.A. (2018). Schaum's Outline of Theory and Problems of Electric Circuits. Seventh edition. McGraw-Hill
Ref08	Siskind, C. (1965). Electrical Circuits: Direct and Alternating Current. Second edition. McGraw-Hill

PART XI. CLASSROOM POLICIES

1. **Attendance and Punctuality**. Classroom attendance is an integral part of the educational experience; therefore, it is the student's responsibility to attend class. Attendance will be checked at every meeting. A student who accumulates exceeding 20% of the prescribed number of class hours will automatically be dropped from the course (unexcused absences). Arriving 15 minutes after the scheduled start time constitutes a late mark. Three late marks equate to one unexcused absence.
2. **Behavior and Decorum**. Respectful, inclusive, and professional behavior is always expected. The use of polite and non-discriminatory language is strictly observed in line with the University's commitment to a gender-fair and inclusive learning environment, and adherence to RA 11313, also known as the Safe Spaces Act.
3. **Gadget Use**. Mobile phones, tablets, and laptops must be stowed away during standard lectures to minimize distractions. However, laptops are expressively permitted and encouraged during designated practical sessions, such as when developing scripts for laboratory data automation or running circuit simulations.

4. **Dress Code.** Wearing the prescribed university uniform is a must in accordance with the Student Handbook.
5. **Submission of Coursework**. Extensions and late submissions of requirements will only be considered if done in writing to the lecturer at least two days before the due date. Generally, no extension will be given for the assignments. Please contact your subject adviser if you encounter any problems.
   Note: (Submission / Resubmission):
   - *Should work be submitted after the due date without written approval for an extension, a penalty of 5% deduction from the grade will be incurred for every 2 days it is late.*
   - *Should the work be assessed as being close to a PASS grade, but the standard has not yet been fully met, the lecturer may offer the opportunity to resubmit.*
6. **Use of AI Tools**. The responsible use of generative AI tools (e.g., ChatGPT, Grammarly, Quillbot) is permitted only when explicitly allowed by the instructor for specific tasks. All AI-assisted work must be properly cited or declared.
7. **Academic Honesty**. All USePians are bound to follow the rules and regulations stipulated in the student handbook of the University and the (College/School). All University policies regarding academic honesty apply to this course. Academic dishonesty includes, but is not limited to, cheating in any form, plagiarism, and similar offenses.
   - Cheating is a serious offense. If you cheat on any assignment or examination, the minimum penalty is a grade of zero for that work. Your conduct may also be referred to in the Office of Student Affairs and Services (OSAS), program head, and dean for appropriate sanctions under university policy.
   - Students are encouraged to work together in exchanging ideas and discussing their assignments in general. However, it is also expected that the work of analyzing data and writing reports will be done in their respective groups. If reports or data have been copied from another student, both students will receive a mark of zero on that examination, test, report, or assignment.
   - Submitting AI-generated content as original work without disclosure is considered academic dishonesty. The acceptable level of matched content using university-accepted plagiarism detection software is 15%. Any paper exceeding this threshold may be subject to rejection or revision.
8. **Conduct of Quizzes and Examinations.** Every quiz and exam, students must follow the general rules:
   - Keep the jacket of their calculators, smartphones, and other things irrelevant to the quiz or exam in their respective bags. If the student caught with any of those objects during the quiz and/or examination proper, it would serve as cheating and dealt accordingly in accordance with the policies imposed by the latter.
   - Instructions must be observed with utmost compliance. Unable to do so will lead to consequences (e.g., deduction of points or answers will not be recognized). During the solution part, rounding up is not recommended as it will accumulate rounding-off errors to the final answer.
   - Write answers legibly using a black or blue ballpoint pen. Show all pertinent solutions clearly; no credit will be given to final answers without detailed, logical steps in accordance with the rubrics. Final answers must be in three decimal places.
   - Math plate is observed to be used during quizzes and bluebook for term examinations. In case the former is unavailable, bluebook can be used as a substitute.
   - No special quizzes or examinations will be given except for valid, documented reasons (e.g., medical certificate, communication letter to attend an activity)
9. **Use of Calculators.** Allowed calculators must follow the Professional Regulatory Commission (PRC) recommendations, see https://www.prc.gov.ph/node/6984 for the full list of non-programmable calculators allowed by the Commission.
10. **Group Work and Independent Effort:** Every group must ensure equitable contributions from all members. While collaboration between different groups for conceptual problem-solving is encouraged, directly sharing completed circuits, code, or written reports are considered plagiarism and will result in a zero (0) for all involved parties.
11. **Cleanliness and Energy Conservation**. Observe cleanliness. Before or after the class, ensure the classroom is clean and dispose of any trash properly. After each class session, ensure that everything is in order and all plugged devices and lights are turned off before leaving the classroom.
12. **Communication Channels:** Official announcements, schedules, and course updates will be posted via the USeP Virtual Environment (UVE), email(s), or the official class group chat. Students are expected to check these platforms regularly. Direct communication with the instructor (via email, messenger, text, or calls) must be conducted strictly during official working days and hours (Monday to Friday, 8:00 AM to 5:00 PM). Any messages, inquiries, or emails sent outside of these designated hours, on weekends, or during holidays will be acknowledged and addressed on the next working day. The only exemption is for messages of immediate urgency, which may be acknowledged outside of these standard hours.

Prepared by:	**Reviewed by:**	**Approved:**
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